



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 29.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : CE G565
Course Title : **Transportation Planning**
Instructor-in-Charge : Dr. Durgesh Vikram (Email: durgesh.vikram@pilani.bits-pilani.ac.in,
Chamber number: 1211 C)
Instructor(s) : NA
Tutorial/Practical Instructors: NA

1. Course Description: Basic concepts in transportation planning, accessibility and mobility, land use interaction, government role in transportation planning, characteristics of travel and transport problems; transportation survey and data collection: planning, design and implementation, travel analysis zone (TAZ) development, traditional four-step modelling process; analysis of travel behaviour and demand: studying travel behaviour, analysing urban travel markets; innovations in transportation modelling: travel behaviour model, activity-based models, econometric modelling using r-studio, modelling travel demand with cube, transportation demand management (TDM), transportation system management (TSM), smart city transportation planning; transit-oriented development (TOD), pedestrian-oriented development, liveable street planning, multimodal transportation planning, shared mobility concepts, integrated transportation management and planning, transportation and energy, climate change, fuel choice and green mobility.

2. Scope and Objective of the Course: The course aims at imparting knowledge on understanding of urban transportation problems from planners' perspective, definition of the problem, setting clear goals and objectives to serve as guiding factors in the planning process, identification of the causal factors influencing the demand for urban travel and development of relationship between the factors and the travel demand.

3. Course learning outcomes:

- a. The candidates will understand the process of planning transportation infrastructures,
- b. . The candidates will learn to model and analyze demand of transportation infrastructures.

4. Text Books:

TB1. Sarkar, P.K., Maitri, V., and Joshi, G.J. *Transportation Planning, Principles, Practices and Policies*, PHI Pvt. Ltd., latest edition.

TB2. Chakroborty, P. and Das, A. *Principles of Transportation Engineering*, PHI Pvt. Ltd., latest edition.

RB1. Papacosta, C.S., and Prevedouros *Transportation Engineering and Planning*, PHI Pvt. Ltd., latest edition.

RB2. Kadiyali, L. R. *Traffic Engineering and Transport Planning*, Khanna Publishers, latest edition.

5. [Method of conduct of the course]: By using both slides (using Power Point) and, chalk and board.



6. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1	5 lectures on transportation planning process and basic concepts of travel demand.	T1 and T2	To understand the transportation planning process.
2	9 lectures on travel surveys and data collection, and travel demand estimation; this will include trip generation and trip distribution	T1	To learn data collection procedures for collecting travel demand data. Also, to understand estimation techniques for trip generation and trip distribution
3	6 lectures on modal share estimation using Utility Theory and Multinomial logit model	T1 and T2	To understand the experiment design for choice modeling to quantify the modal share of competing modes.
4	8 lectures on traffic assignment techniques using various shortest path algorithms	T1 and T2	To understand different algorithms to find the shortest path and learn traffic assignment techniques.
5	4 lectures on regional as well as long-term transportation planning	T1	To understand the regional planning process and the fundamentals of preparing long-term transport plans.
6	4 lectures on transportation systems management	R3	To learn the management strategies for managing the travel demand.

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	25	Refer to the timetable	Closed Book
Comprehensive Examination	3 h	35	Refer to the timetable	Closed Book
Quiz/Assignment		20	Spread over the semester	Open/Closed Book
Term paper		20	Spread over the semester	Open Book

7. Criterion for NC: 30% of the average of the top three performers or 40% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.

8. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

9. Chamber consultation hours: To be announced during the lecture.

10. Makeup Policy: Makeup will be given only for genuine cases (accompanied by proof) for Mid-sem or comprehensive exam only.

Instructor-in-charge
Course No. CE G565